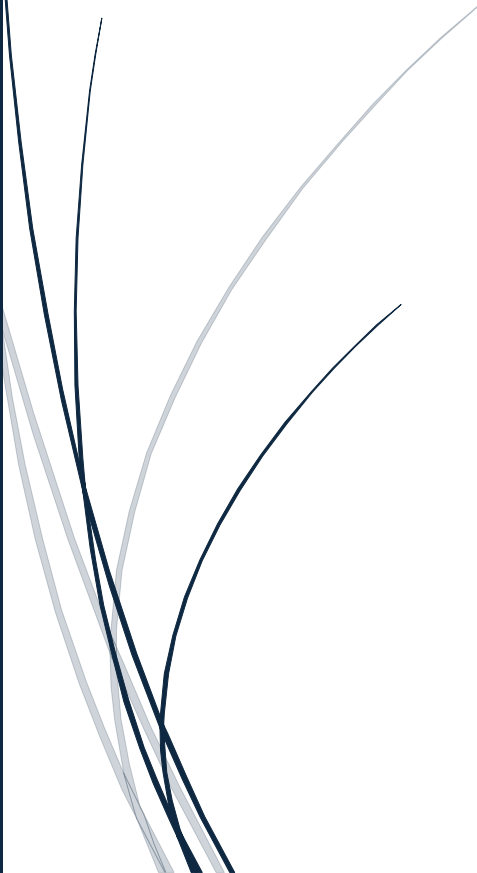




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Network security

Multi-Class Network Intrusion
Detection System (IDS) for Flood
Attacks using Random Forest



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Introduction and Project Goal

The project begins by establishing the need for an **Intrusion Detection System (IDS)**, a critical security mechanism used to monitor and analyze network traffic for patterns indicating malicious activity or policy violations. The primary goal of this project is to build a **Multiclass Classification** machine learning model capable of distinguishing between three core traffic patterns:

1. Normal Traffic (Safe and legitimate network activity).
2. ICMP Flood Attack (A Denial-of-Service attack).
3. SYN Flood Attack (A resource exhaustion attack).

Targeted Attacks:

1. ICMP Flood Attack

This is one of the simplest and most widespread Denial-of-Service (DoS) attacks. It is executed by overwhelming the victim with a high rate of **ICMP Echo Request** packets (similar to the Ping command). The attack's objective is to consume the victim's **bandwidth** and CPU resources, as the victim is forced to send an **Echo Reply** for every request, ultimately preventing legitimate connections from being processed.

2. SYN Flood Attack

This attack exploits the three-way handshake mechanism of the TCP protocol. The attacker sends a massive number of **SYN** packets (connection requests) but never responds to the victim's **SYN-ACK** packets. This maneuver leaves a large number of connections in a **"Half-Open"** state in the victim's connection queue. This consumes the system's memory resources (the connection table) and hinders the server's ability to accept legitimate connection requests.

Lab Setup and Topology

The entire project was executed within an **Isolated Virtual Environment** to ensure the attack simulations did not interfere with the main network infrastructure. The setup involved two primary virtual machines connected within a private, internal network, simulating a direct attack scenario in a controlled setting.

Key Components and Configuration:

Environment: The entire lab utilized an isolated virtual network topology.

Attacker and IDS Host (Kali Linux):

- Served as the platform for executing the simulated attacks (using hping3).
- Hosted the training environment for the Machine Learning model (Python, Scikit-learn, Scapy).
- Ran the final *Intrusion Detection System (IDS)* monitor.

Victim Host (Metasploitable 2):

- Represented the vulnerable target server for the Denial-of-Service (DoS) and SYN Flood attacks.
- Specific open ports (e.g., Port 80) were targeted.

Data Capture Tool (Tshark / Wireshark):

- Used for passively capturing and extracting the raw packet data necessary for feature engineering and dataset creation.

Network Addressing :

- The systems were configured on a private subnet (Attacker IP: 192.168.241.129, Target IP: 192.168.241.131).

Attack Execution Steps and Commands

Traffic data collection involved running three distinct scenarios on the Kali Linux machine against the Metasploitable 2 victim, ensuring clear separation for dataset labeling.

ICMP Flood Attack

- This phase aimed to generate a high volume of ICMP Echo Request packets to simulate a basic Denial-of-Service condition.
- Objective: To overwhelm the victim's resources and bandwidth by flooding it with ICMP packets.
- Execution Command: The attack was executed using the following single-line command: **sudo hping3 -1 --flood <Target_IP>**
- Command Breakdown: The -1 flag specifies the ICMP protocol, and the --flood flag instructs the tool to send packets as fast as possible without waiting for replies.

```
(root@kali)-[~]
# sudo hping3 --icmp --flood -d 120 192.168.241.131
HPING 192.168.241.131 (eth0 192.168.241.131): icmp mode set, 28 headers + 120 data bytes
hping in flood mode, no replies will be shown
^C
— 192.168.241.131 hping statistic — 192.168.241.131: 0>1: 192.168.241.131
104411 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
(root@kali)-[~]
#
```

SYN Flood Attack

- This attack targeted a specific open TCP port to exhaust the victim's connection resources by leaving connections half-open.
- Objective: To simulate a resource exhaustion attack by exploiting the TCP three-way handshake mechanism.
- Target Identification: Nmap was initially used to confirm that a service port, such as Port 80 (HTTP), was accessible and listening on the victim machine.
- Execution Command: The attack was launched against the identified port: **sudo hping3 -S --flood <Target_IP> -p 80**
- Command Breakdown: The -S flag sets the SYN flag (connection request), and -p 80 specifies the target port. The --flood flag maximizes the attack rate.

```
File Actions Edit View Help
root@kali: ~
# nmap 192.168.241.131
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-05 13:05 EST
Nmap scan report for 192.168.241.131
Host is up (0.067s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:4E:7E:FB (VMware)

Nmap done: 1 IP address (1 host up) scanned in 2.81 seconds
```

```

(root@kali)-[~]
# sudo hping3 -S --flood 192.168.241.131 -p 80
HPING 192.168.241.131 (eth0 192.168.241.131): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
— 192.168.241.131 hping statistic —
251871 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

```

Normal Traffic Generation

- To establish a diverse safe baseline, traffic was generated using natural human interaction patterns (Ping and FTP), ensuring variable timing.

Ping Execution: ping <Target_IP>

- FTP Execution (TCP/Port 21): Simple FTP connections (e.g., login, listing directories) were simulated to record normal TCP traffic.

FTP Execution: ftp <Target_IP>

```

# ping 192.168.241.131
PING 192.168.241.131 (192.168.241.131) 56(84) bytes of data.
64 bytes from 192.168.241.131: icmp_seq=1 ttl=64 time=0.845 ms
64 bytes from 192.168.241.131: icmp_seq=2 ttl=64 time=3.59 ms
64 bytes from 192.168.241.131: icmp_seq=3 ttl=64 time=1.68 ms
64 bytes from 192.168.241.131: icmp_seq=4 ttl=64 time=1.01 ms
64 bytes from 192.168.241.131: icmp_seq=5 ttl=64 time=15.7 ms
64 bytes from 192.168.241.131: icmp_seq=6 ttl=64 time=1.87 ms
64 bytes from 192.168.241.131: icmp_seq=7 ttl=64 time=9.23 ms
64 bytes from 192.168.241.131: icmp_seq=8 ttl=64 time=1.26 ms
64 bytes from 192.168.241.131: icmp_seq=9 ttl=64 time=2.08 ms
64 bytes from 192.168.241.131: icmp_seq=10 ttl=64 time=2.21 ms
64 bytes from 192.168.241.131: icmp_seq=11 ttl=64 time=0.795 ms
64 bytes from 192.168.241.131: icmp_seq=12 ttl=64 time=45.0 ms
64 bytes from 192.168.241.131: icmp_seq=13 ttl=64 time=80.5 ms
64 bytes from 192.168.241.131: icmp_seq=14 ttl=64 time=0.565 ms
^C64 bytes from 192.168.241.131: icmp_seq=15 ttl=64 time=36.8 ms

— 192.168.241.131 ping statistics —
15 packets transmitted, 15 received, 0% packet loss, time 14105ms
rtt min/avg/max/mdev = 0.565/13.540/80.459/22.310 ms

(root@kali)-[~]
# ftp 192.168.241.131
Connected to 192.168.241.131.
220 (vsFTPd 2.3.4)
Name (192.168.241.131:root): msfadmin
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
ftp>
ftp>
ftp>
ftp>
ftp> exit
221 Goodbye.

```

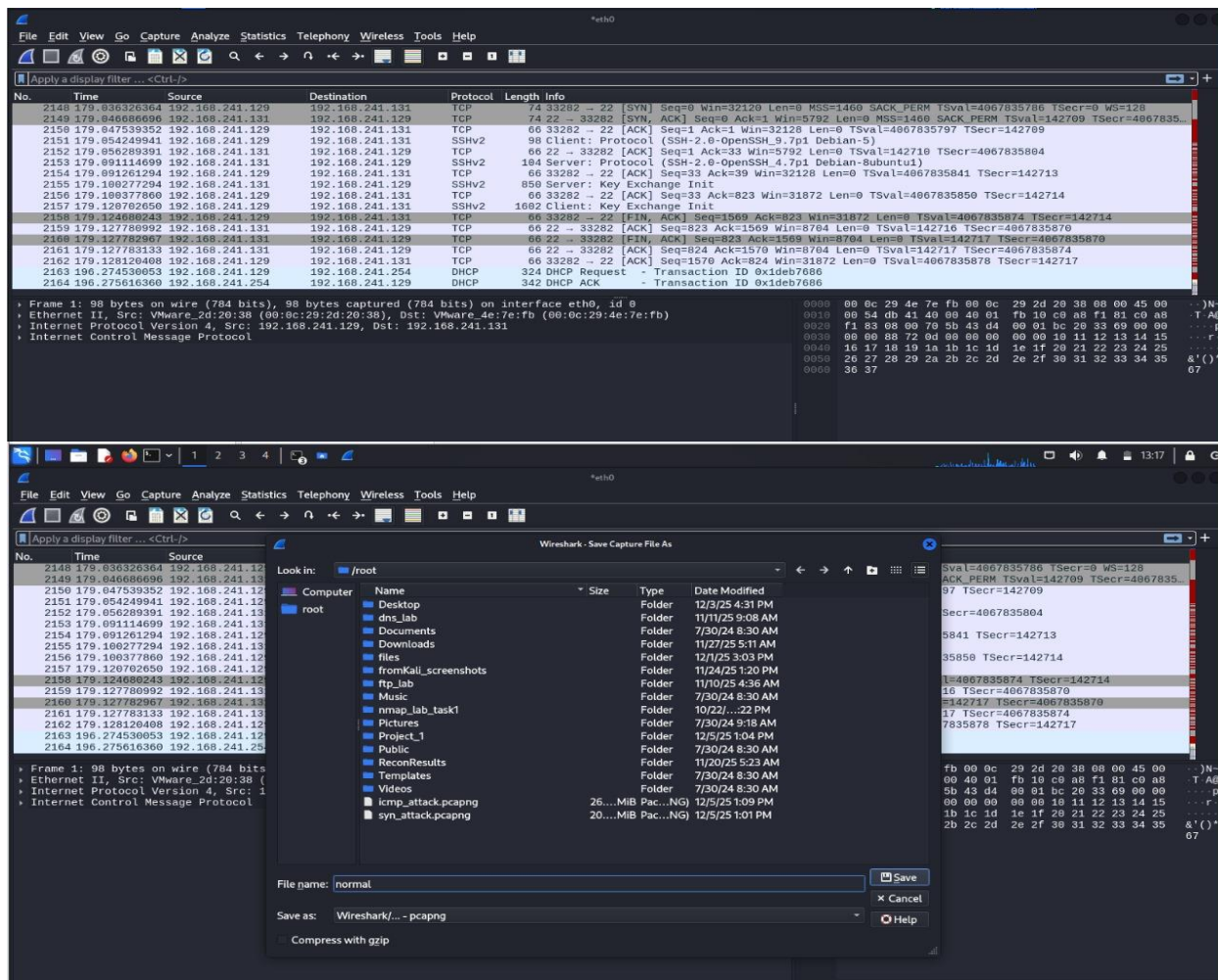

Dataset Creation and Record Counts

The foundation of the IDS model relies on a unified dataset created by processing the raw network captures from the three simulated scenarios (Normal, ICMP Flood, SYN Flood). This process involved using Tshark to extract specific features from the raw packet files (.pcap) and then merging these outputs into a single, cohesive dataset.

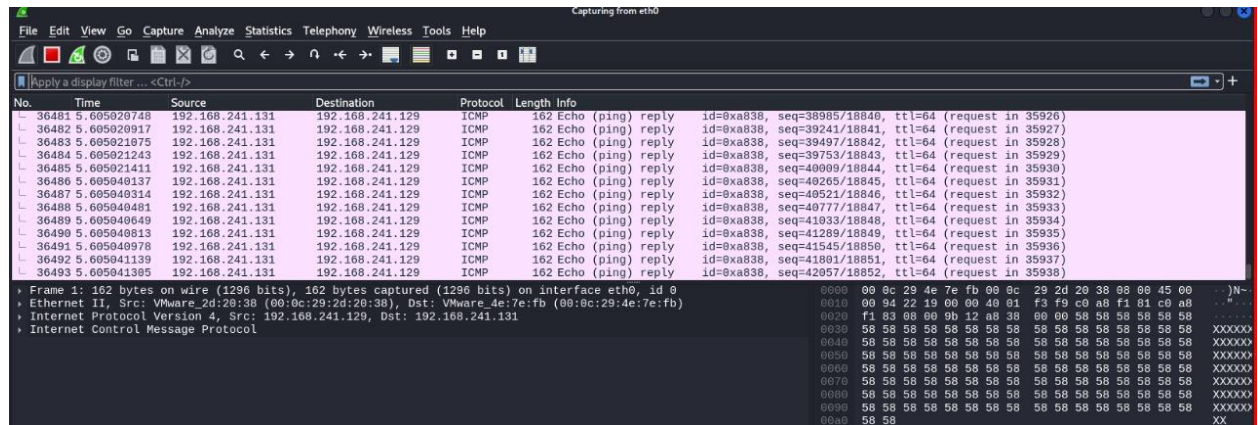
Data Extraction and Preprocessing

- Capture and Labeling: Three separate packet capture files (.pcap) were recorded, each corresponding to a specific label:

Normal Traffic (Label 0)



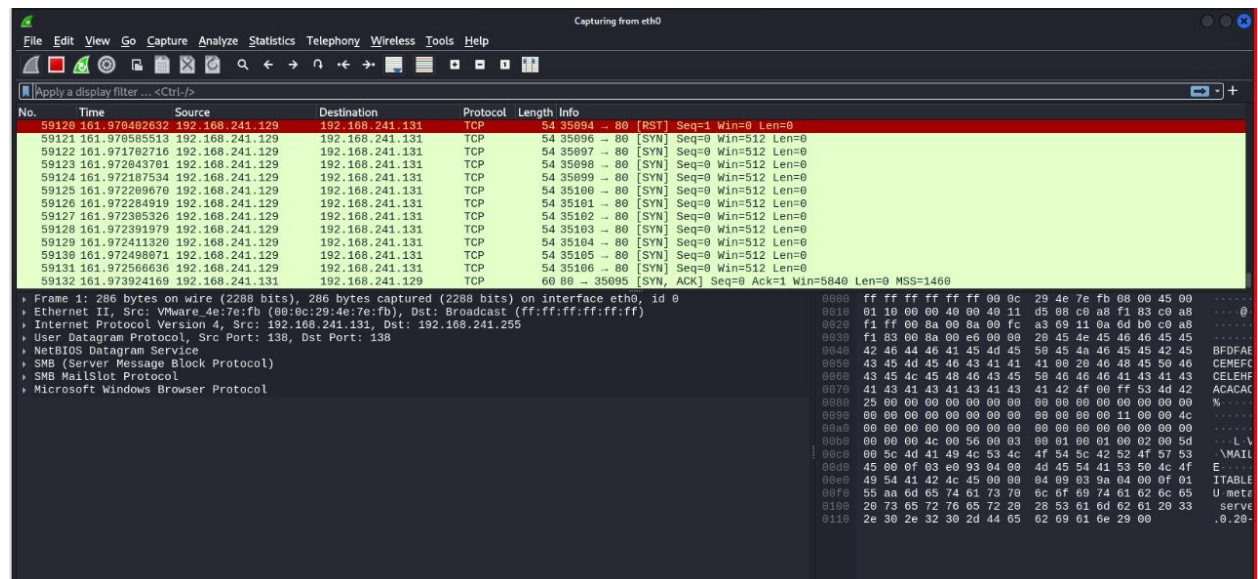
ICMP Flood Attack (Label 1)



The image shows a Wireshark packet capture of an ICMP Flood Attack. The interface includes a menu bar (File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help), a toolbar, and a display filter set to 'No display filter'. The packet list pane shows 13 packets, all of which are ICMP Echo (ping) requests from source IP 192.168.241.131 to destination IP 192.168.241.129. The packet details pane for the selected packet (No. 13) shows the Ethernet II header, Internet Protocol Version 4 header, and Internet Control Message Protocol header. The packet bytes pane displays the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
13	36481	5.605020748	192.168.241.131	ICMP	162	Echo (ping) reply
14	36482	5.605020917	192.168.241.131	ICMP	162	Echo (ping) reply
15	36483	5.605021075	192.168.241.131	ICMP	162	Echo (ping) reply
16	36484	5.605021243	192.168.241.131	ICMP	162	Echo (ping) reply
17	36485	5.605021411	192.168.241.131	ICMP	162	Echo (ping) reply
18	36486	5.605040137	192.168.241.131	ICMP	162	Echo (ping) reply
19	36487	5.605040314	192.168.241.131	ICMP	162	Echo (ping) reply
20	36488	5.605040481	192.168.241.131	ICMP	162	Echo (ping) reply
21	36489	5.605040649	192.168.241.131	ICMP	162	Echo (ping) reply
22	36490	5.605040813	192.168.241.131	ICMP	162	Echo (ping) reply
23	36491	5.605040978	192.168.241.131	ICMP	162	Echo (ping) reply
24	36492	5.605041139	192.168.241.131	ICMP	162	Echo (ping) reply
25	36493	5.605041395	192.168.241.131	ICMP	162	Echo (ping) reply

SYN Flood Attack (Label 2)



The image shows a Wireshark packet capture of a SYN Flood Attack. The interface includes a menu bar (File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help), a toolbar, and a display filter set to 'No display filter'. The packet list pane shows 13 packets, all of which are TCP SYN requests from source IP 192.168.241.131 to destination IP 192.168.241.129. The packet details pane for the selected packet (No. 13) shows the Ethernet II header, Internet Protocol Version 4 header, and User Datagram Protocol header. The packet bytes pane displays the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
59128	161.970402632	192.168.241.129	192.168.241.131	TCP	54	35094 → 80 [RST] Seq=1 Win=0 Len=0
59121	161.970585513	192.168.241.129	192.168.241.131	TCP	54	35096 → 80 [SYN] Seq=0 Win=512 Len=0
59122	161.971027716	192.168.241.129	192.168.241.131	TCP	54	35097 → 80 [SYN] Seq=0 Win=512 Len=0
59123	161.972043701	192.168.241.129	192.168.241.131	TCP	54	35098 → 80 [SYN] Seq=0 Win=512 Len=0
59124	161.972187534	192.168.241.129	192.168.241.131	TCP	54	35099 → 80 [SYN] Seq=0 Win=512 Len=0
59125	161.972289670	192.168.241.129	192.168.241.131	TCP	54	35100 → 80 [SYN] Seq=0 Win=512 Len=0
59126	161.972284919	192.168.241.129	192.168.241.131	TCP	54	35101 → 80 [SYN] Seq=0 Win=512 Len=0
59127	161.972305326	192.168.241.129	192.168.241.131	TCP	54	35102 → 80 [SYN] Seq=0 Win=512 Len=0
59128	161.972391979	192.168.241.129	192.168.241.131	TCP	54	35103 → 80 [SYN] Seq=0 Win=512 Len=0
59129	161.972411326	192.168.241.129	192.168.241.131	TCP	54	35104 → 80 [SYN] Seq=0 Win=512 Len=0
59130	161.972498071	192.168.241.129	192.168.241.131	TCP	54	35105 → 80 [SYN] Seq=0 Win=512 Len=0
59131	161.972566636	192.168.241.129	192.168.241.131	TCP	54	35106 → 80 [SYN] Seq=0 Win=512 Len=0
59132	161.973924169	192.168.241.131	192.168.241.129	TCP	60	80 → 35095 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460

- **Feature Extraction via Tshark:** The captured files were processed using the command-line utility Tshark. The tool was instructed to extract the 9 features deemed most critical for classification, ensuring maximum separation between the classes.
- **Command Structure:** The robust command used -T fields and -E separator=, to generate clean CSV output, converting complex packet data into easily readable numerical features.
- **Data Unification and Cleaning:** The three resulting CSV files were loaded into a Python script using Pandas. A specific label (0, 1, or 2) was

assigned to each file's records, and all three datasets were merged into a single file: final_multiclass_dataset.csv.

```
(root@kali)~# python3 merge_data.py
/root/merge_data.py:14: DtypeWarning: Columns (5) have mixed types. Specify dtype option on import or set low_memory=False.
df_syn = pd.read_csv(SYN_FILE)
Files loaded successfully: Normal, ICMP, and SYN.

Success! Multiclass dataset created: 'final_multiclass_dataset.csv'
Total Records: 360476
Normal (0) Records: 2164
ICMP (1) Records: 140163
SYN (2) Records: 218149
```

```
Warning: you are using the root account. You may harm your system.

1|frame.time_delta,ip.proto,ip.len,tcp.srcport,tcp.dstport,tcp.flags.syn,icmp.type,icmp.seq,icmp.code,Label
2 0.0,1.0,84.0,0.0,0.0,0.0,8.0,1.0,0.0,0
3 0.000807785,1.0,84.0,0.0,0.0,0.0,8.0,1.0,0.0,0
4 1.003065824,1.0,84.0,0.0,0.0,0.0,8.0,2.0,0.0,0
5 0.003505926,1.0,84.0,0.0,0.0,0.0,8.0,2.0,0.0,0
6 0.998944461,1.0,84.0,0.0,0.0,0.0,8.0,3.0,0.0,0
7 0.001552472,1.0,84.0,0.0,0.0,0.0,8.0,3.0,0.0,0
8 1.00006209,1.0,84.0,0.0,0.0,0.0,8.0,4.0,0.0,0
9 0.000896582,1.0,84.0,0.0,0.0,0.0,8.0,4.0,0.0,0
10 1.002301956,1.0,84.0,0.0,0.0,0.0,8.0,5.0,0.0,0
11 0.015702194,1.0,84.0,0.0,0.0,0.0,8.0,5.0,0.0,0
12 0.970242944,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0
13 4.9304e-05,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0
14 0.01613319,1.0,84.0,0.0,0.0,0.0,8.0,6.0,0.0,0
15 0.00180497,1.0,84.0,0.0,0.0,0.0,8.0,6.0,0.0,0
16 0.050140049,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0
17 0.000845559,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0
18 0.950229322,1.0,84.0,0.0,0.0,0.0,8.0,7.0,0.0,0
19 0.009154835,1.0,84.0,0.0,0.0,0.0,8.0,7.0,0.0,0
20 0.992010977,1.0,84.0,0.0,0.0,0.0,8.0,8.0,0.0,0
21 0.001103518,1.0,84.0,0.0,0.0,0.0,8.0,8.0,0.0,0
22 1.001085312,1.0,84.0,0.0,0.0,0.0,8.0,9.0,0.0,0
23 0.002011916,1.0,84.0,0.0,0.0,0.0,8.0,9.0,0.0,0
24 1.001291609,1.0,84.0,0.0,0.0,0.0,8.0,10.0,0.0,0
25 0.002123294,1.0,84.0,0.0,0.0,0.0,8.0,10.0,0.0,0
26 0.999992896,1.0,84.0,0.0,0.0,0.0,8.0,11.0,0.0,0
27 0.000701221,1.0,84.0,0.0,0.0,0.0,8.0,11.0,0.0,0
28 1.017332738,1.0,84.0,0.0,0.0,0.0,8.0,12.0,0.0,0
29 0.044913307,1.0,84.0,0.0,0.0,0.0,8.0,12.0,0.0,0
30 0.957379762,1.0,84.0,0.0,0.0,0.0,8.0,13.0,0.0,0
31 0.080394746,1.0,84.0,0.0,0.0,0.0,8.0,13.0,0.0,0
32 0.92104648,1.0,84.0,0.0,0.0,0.0,8.0,14.0,0.0,0
33 0.000516002,1.0,84.0,0.0,0.0,0.0,8.0,14.0,0.0,0
34 1.057085353,1.0,84.0,0.0,0.0,0.0,8.0,15.0,0.0,0
35 0.030731371,1.0,84.0,0.0,0.0,0.0,8.0,15.0,0.0,0
36 12.304773972,6.0,60.0,21.0,34996.0,21.0,True,0.0,0.0,0
37 0.001089612,6.0,60.0,21.0,34996.0,True,0.0,0.0,0.0,0
38 0.000170364,6.0,52.0,34996.0,21.0,False,0.0,0.0,0.0,0
```

```
Warning: you are using the root account. You may harm your system.

31197 3.20726e-05,1.0,148.0,0.0,0.0,0.0,8.0,10099.0,0.0,1
31198 0.000203084,1.0,148.0,0.0,0.0,0.0,8.0,16955.0,0.0,1
31199 2.9616e-05,1.0,148.0,0.0,0.0,0.0,8.0,17211.0,0.0,1
31200 7.5986e-05,1.0,148.0,0.0,0.0,0.0,8.0,17467.0,0.0,1
31201 1.8854e-05,1.0,148.0,0.0,0.0,0.0,8.0,17723.0,0.0,1
31202 7.957e-05,1.0,148.0,0.0,0.0,0.0,8.0,17979.0,0.0,1
31203 0.000147484,1.0,148.0,0.0,0.0,0.0,8.0,18235.0,0.0,1
31204 0.00053431,1.0,148.0,0.0,0.0,0.0,8.0,18491.0,0.0,1
31205 5.3817e-05,1.0,148.0,0.0,0.0,0.0,8.0,18747.0,0.0,1
31206 0.000654773,1.0,148.0,0.0,0.0,0.0,8.0,19003.0,0.0,1
31207 2.8901e-05,1.0,148.0,0.0,0.0,0.0,8.0,19259.0,0.0,1
31208 -2.325e-06,1.0,148.0,0.0,0.0,0.0,8.0,14651.0,0.0,1
31209 6.24e-07,1.0,148.0,0.0,0.0,0.0,8.0,14907.0,0.0,1
31210 1.79e-07,1.0,148.0,0.0,0.0,0.0,8.0,15163.0,0.0,1
31211 1.71e-07,1.0,148.0,0.0,0.0,0.0,8.0,15419.0,0.0,1
31212 1.77e-07,1.0,148.0,0.0,0.0,0.0,8.0,15675.0,0.0,1
31213 1.74e-07,1.0,148.0,0.0,0.0,0.0,8.0,15931.0,0.0,1
31214 1.68e-07,1.0,148.0,0.0,0.0,0.0,8.0,16187.0,0.0,1
31215 1.69e-07,1.0,148.0,0.0,0.0,0.0,8.0,16443.0,0.0,1
31216 0.000283997,1.0,148.0,0.0,0.0,0.0,8.0,16699.0,0.0,1
31217 6.73e-07,1.0,148.0,0.0,0.0,0.0,8.0,16955.0,0.0,1
31218 1.99e-07,1.0,148.0,0.0,0.0,0.0,8.0,17211.0,0.0,1
31219 1.9e-07,1.0,148.0,0.0,0.0,0.0,8.0,17467.0,0.0,1
31220 3.6e-07,1.0,148.0,0.0,0.0,0.0,8.0,17723.0,0.0,1
31221 1.76e-07,1.0,148.0,0.0,0.0,0.0,8.0,17979.0,0.0,1
31222 3.34e-07,1.0,148.0,0.0,0.0,0.0,8.0,18235.0,0.0,1
31223 0.000246348,1.0,148.0,0.0,0.0,0.0,8.0,19515.0,0.0,1
31224 4.3416e-05,1.0,148.0,0.0,0.0,0.0,8.0,19771.0,0.0,1
31225 0.000201872,1.0,148.0,0.0,0.0,0.0,8.0,20027.0,0.0,1
31226 2.3198e-05,1.0,148.0,0.0,0.0,0.0,8.0,20283.0,0.0,1
31227 0.000223116,1.0,148.0,0.0,0.0,0.0,8.0,20539.0,0.0,1
31228 2.3358e-05,1.0,148.0,0.0,0.0,0.0,8.0,20795.0,0.0,1
31229 0.000159519,1.0,148.0,0.0,0.0,0.0,8.0,21051.0,0.0,1
31230 2.2584e-05,1.0,148.0,0.0,0.0,0.0,8.0,21307.0,0.0,1
31231 0.000199014,1.0,148.0,0.0,0.0,0.0,8.0,21563.0,0.0,1
31232 4.4907e-05,1.0,148.0,0.0,0.0,0.0,8.0,21819.0,0.0,1
31233 8.6528e-05,1.0,148.0,0.0,0.0,0.0,8.0,22075.0,0.0,1
31234 2.1156e-05,1.0,148.0,0.0,0.0,0.0,8.0,22331.0,0.0,1
-----
```

```
Warning: you are using the root account. You may harm your system.

143697 5.7287e-05,6.0,44.0,58020.0,9091.0,True,0.0,0.0,0.0,2
143698 0.000123527,6.0,44.0,58020.0,2608.0,True,0.0,0.0,0.0,2
143699 5.9137e-05,6.0,44.0,58020.0,32779.0,True,0.0,0.0,0.0,2
143700 2.507e-06,6.0,40.0,14441.0,58020.0,False,0.0,0.0,0.0,2
143701 2.93e-07,6.0,40.0,4111.0,58020.0,False,0.0,0.0,0.0,2
143702 5.2454e-05,6.0,44.0,58020.0,12345.0,True,0.0,0.0,0.0,2
143703 5.7089e-05,6.0,44.0,58020.0,8654.0,True,0.0,0.0,0.0,2
143704 0.000250655,6.0,40.0,3801.0,58020.0,False,0.0,0.0,0.0,2
143705 5.2541e-05,6.0,44.0,58020.0,4005.0,True,0.0,0.0,0.0,2
143706 0.000588045,6.0,40.0,6839.0,58020.0,False,0.0,0.0,0.0,2
143707 2.6885e-05,6.0,44.0,58020.0,2125.0,True,0.0,0.0,0.0,2
143708 0.001385321,6.0,40.0,9091.0,58020.0,False,0.0,0.0,0.0,2
143709 9.32e-07,6.0,40.0,2608.0,58020.0,False,0.0,0.0,0.0,2
143710 1.64e-07,6.0,40.0,32779.0,58020.0,False,0.0,0.0,0.0,2
143711 1.63e-07,6.0,40.0,12345.0,58020.0,False,0.0,0.0,0.0,2
143712 1.64e-07,6.0,40.0,8654.0,58020.0,False,0.0,0.0,0.0,2
143713 1.05e-07,6.0,40.0,4005.0,58020.0,False,0.0,0.0,0.0,2
143714 1.07e-07,6.0,40.0,2125.0,58020.0,False,0.0,0.0,0.0,2
143715 3.5481e-05,6.0,44.0,58020.0,50636.0,True,0.0,0.0,0.0,2
143716 0.000623296,6.0,44.0,58020.0,5226.0,True,0.0,0.0,0.0,2
143717 0.000825697,6.0,40.0,50636.0,58020.0,False,0.0,0.0,0.0,2
143718 1.489e-06,6.0,40.0,5226.0,58020.0,False,0.0,0.0,0.0,2
143719 0.000325357,6.0,44.0,58020.0,2000.0,True,0.0,0.0,0.0,2
143720 0.000638990,6.0,40.0,2000.0,58020.0,False,0.0,0.0,0.0,2
143721 5.8676e-05,6.0,44.0,58020.0,9595.0,True,0.0,0.0,0.0,2
143722 0.000279843,6.0,44.0,58020.0,8873.0,True,0.0,0.0,0.0,2
143723 0.000274566,6.0,40.0,9595.0,58020.0,False,0.0,0.0,0.0,2
143724 4.11e-07,6.0,40.0,8873.0,58020.0,False,0.0,0.0,0.0,2
143725 0.000280053,6.0,44.0,58020.0,9200.0,True,0.0,0.0,0.0,2
143726 0.000284484,6.0,44.0,58020.0,1300.0,True,0.0,0.0,0.0,2
143727 7.2168e-05,6.0,40.0,9200.0,58020.0,False,0.0,0.0,0.0,2
143728 0.000286956,6.0,44.0,58020.0,8443.0,True,0.0,0.0,0.0,2
143729 3.9326e-05,6.0,40.0,1300.0,58020.0,False,0.0,0.0,0.0,2
143730 1.4499e-05,6.0,44.0,58020.0,3945.0,True,0.0,0.0,0.0,2
143731 7.4503e-05,6.0,44.0,58020.0,2383.0,True,0.0,0.0,0.0,2
143732 1.3156e-05,6.0,44.0,58020.0,9929.0,True,0.0,0.0,0.0,2
143733 0.000202152,6.0,44.0,58020.0,5904.0,True,0.0,0.0,0.0,2
143734 8.4289e-05,6.0,44.0,58020.0,19.0,True,0.0,0.0,0.0,2
```

Final Record Counts

The final merged dataset contained sufficient records to ensure the model was exposed to a large variety of traffic patterns. The data was split into training and testing sets to validate the model's ability to generalize its learning.

- Total Dataset Size: The overall collection of processed and labeled packets amounted to 360,476 records.
- Training Set (70%): 252,333 records were allocated for training the Random Forest model to learn the attack and normal patterns.
- Testing Set (30%): 108,143 records were reserved for the final evaluation to verify the model's accuracy on unseen data.
- Breakdown by Classification Class:

Normal Traffic (Label 0): Approximately 651 records were included to establish the safe network baseline.

ICMP Flood Attack (Label 1): Approximately 42,002 records were generated to represent the ICMP denial-of-service attack pattern.

SYN Flood Attack (Label 2): Approximately 65,490 records were collected, representing the SYN connection exhaustion attack pattern.

```
(root@kali) ~# python3 ai_model.py
Loading dataset from final_multiclass_dataset.csv...
/root/ai_model.py:22: DtypeWarning: Columns (5) have mixed types. Specify dtype option on import or set low_memory=False.
df = pd.read_csv(DATA_FILE)
Training set size: 252333
Test set size: 108143

Training Random Forest Multiclass Model...
Evaluating ...

Model Accuracy: 100.00%

Classification Report (0=Normal, 1=ICMP, 2=SYN):

```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	651
1	1.00	1.00	1.00	42002
2	1.00	1.00	1.00	65490
accuracy			1.00	108143
macro avg	1.00	1.00	1.00	108143
weighted avg	1.00	1.00	1.00	108143

```

Confusion Matrix:
[[ 648  1  2]
 [ 0 42001 1]
 [ 1  0 65489]]

Feature Importance (Most influential features):
1. ip.len: 0.3888
2. icmp.seq: 0.2812
3. ip.proto: 0.1951
4. tcp.srcport: 0.0648
5. tcp.dstport: 0.0592
6. icmp.type: 0.0080
7. tcp.flags.syn: 0.0021
8. frame.time_delta: 0.0008
9. icmp.code: 0.0000

Model saved successfully as 'multiclass_ids_model.joblib'
```

Feature Descriptions

The effectiveness of the IDS model depends on selecting features capable of translating attack behavior into numerical values. The final **9 features** used in the model can be categorized by their primary function:

Speed and Size Indicators

- **frame.time_delta (Inter-Packet Arrival Time):** This feature is crucial for detecting floods. A value close to **zero** indicates that packets are being sent at the maximum rate (attack), while larger, variable values signify normal traffic.
- **ip.len (Total IP Length):** This reveals the size of the packet. In flood attacks, this value is often large and uniform, contrasting sharply with the varied sizes found in legitimate traffic.

Protocol Identification and Flagging

- **ip.proto (IP Protocol Type):** A foundational value that distinguishes the core protocol being used (e.g., **1** for ICMP, **6** for TCP).
- **icmp.type and icmp.code:** These fields specifically identify the ICMP packet as an **Echo Request**, which is the precise fingerprint of the ICMP Flood attack.
- **tcp.flags.syn (SYN Flag Status):** A binary value (0 or 1) that indicates if the SYN flag is set. The continuous high-frequency recurrence of '1' is the signature of a SYN Flood.

Sequencing and Connection Context

- **icmp.seq (ICMP Sequence Number):** In a flood, this number increases rapidly and linearly, which clearly exposes the automated, scripted generation of attack packets by tools like hping3.
- **tcp.srcport and tcp.dstport (TCP Ports):** These help establish the context of TCP/SYN traffic and aid in separating port-based attacks (SYN) from protocol-based attacks (ICMP).

```
(root@kali) ~  
# tshark -r normal.pcapng -T fields -e frame.time_delta -e ip.proto -e ip.len -e tcp.srcport -e tcp.dstport -e tcp.flags.syn -e icmp.type -e icmp.seq -e icmp.code -E header=y -E separator=, -E quote=d -E occurrence=f > normal.csv  
Running as user "root" and group "root". This could be dangerous.  
  
(root@kali) ~  
# tshark -r syn_attack.pcapng -T fields -e frame.time_delta -e ip.proto -e ip.len -e tcp.srcport -e tcp.dstport -e tcp.flags.syn -e icmp.type -e icmp.seq -e icmp.code -E header=y -E separator=, -E quote=d -E occurrence=f > syn_attack.csv  
Running as user "root" and group "root". This could be dangerous.  
^C  
  
(root@kali) ~  
# tshark -r icmp_attack.pcapng -T fields -e frame.time_delta -e ip.proto -e ip.len -e tcp.srcport -e tcp.dstport -e tcp.flags.syn -e icmp.type -e icmp.seq -e icmp.code -E header=y -E separator=, -E quote=d -E occurrence=f > icmp_attack.csv  
Running as user "root" and group "root". This could be dangerous.
```

Feature Selection Method & Results

Selection Methodology

- The final selection of features was determined using a Model-Based Feature Selection approach to robustly identify the most predictive features in distinguishing between the three traffic classes (Normal, ICMP, SYN).
- Algorithm Used: The Feature Importance property inherent to the Random Forest Classifier was leveraged.
- Mechanism: This technique calculates how much each feature contributes to reducing the overall data impurity (specifically, Gini Impurity) when the decision trees are constructed. Features that provide the cleanest, quickest split between attack data and normal data are ranked highest. This method is effective for showing the actual operational influence of each packet field on the model's decision-making process.

Feature Importance Results

- The training evaluation confirmed that the model's high accuracy relies heavily on size, core protocol, and sequencing patterns. The top three most influential features, which accounted for over 86% of the model's predictive power, are listed below: ip.len (Total IP Length): 0.3888
- Analysis: This was the single most influential factor. The model learned that the uniformity and large magnitude of the packet size in the flood traffic provided the cleanest data separation, making it the primary discriminator between Normal and Attack traffic.
icmp.seq (ICMP Sequence Number): 0.2812
- Analysis: The second most important feature, providing a strong signal for ICMP Flood detection. The rapid, linear increase in the sequence

number strongly identifies the automated, scripted nature of the attack generation.

ip.proto (IP Protocol Type): 0.1951

- **Analysis:** A foundational feature that immediately separates ICMP traffic (Protocol 1) from TCP/SYN traffic (Protocol 6), which is essential for multi-class distinction.

```
Feature Importance (Most influential features):
1. ip.len: 0.3888
2. icmp.seq: 0.2812
3. ip.proto: 0.1951
4. tcp.srcport: 0.0648
5. tcp.dstport: 0.0592
6. icmp.type: 0.0080
7. tcp.flags.syn: 0.0021
8. frame.time_delta: 0.0008
9. icmp.code: 0.0000

Model saved successfully as 'multiclass_ids_model.joblib'
```

ML Models Used and Explanation

The **Random Forest Classifier algorithm** was selected as the primary model for training the multi-class Intrusion Detection System (IDS).

Justification for Choosing Random Forest

Random Forest is an Ensemble Learning model that builds a large number of decision trees independently, aggregating their results to make the final classification decision. It was chosen for this project for the following reasons:

- **Efficiency in Multiclass Classification:** Random Forest is inherently designed to handle more than two classes, making it ideal for the task of distinguishing between (0=Normal, 1=ICMP, 2=SYN).
- **Robustness in Tabular Data:** Network traffic features are structured as tabular, numerical data, and Random Forest generally excels with this

type of input, providing high performance without requiring extensive data normalization.

- **Resistance to Overfitting:** The model reduces the risk of overfitting by basing its final decision on the consensus of many individual trees rather than relying on a single, potentially over-fitted tree.
- **Built-in Feature Importance:** The model provides a direct measure of **Feature Importance** (which was used and documented in Section 7), simplifying the understanding and interpretation of the model's classification decisions.

```
(root@kali)~# python3 ai_model.py
Loading dataset from final_multiclass_dataset.csv ...
/root/ai_model.py:22: DtypeWarning: Columns (5) have mixed types. Specify dtype option on import or set low_memory=False.
df = pd.read_csv(DATA_FILE)
Training set size: 252333
Test set size: 108143

Training Random Forest Multiclass Model ...
Evaluating ...

Model Accuracy: 100.00%

Classification Report (0=Normal, 1=ICMP, 2=SYN):

```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	651
1	1.00	1.00	1.00	42002
2	1.00	1.00	1.00	65490
accuracy			1.00	108143
macro avg	1.00	1.00	1.00	108143
weighted avg	1.00	1.00	1.00	108143

```

Confusion Matrix:
[[ 648   1   2]
 [  0 42001  1]
 [  1   0 65489]]

Feature Importance (Most influential features):
1. ip.len: 0.3888
2. icmp.seq: 0.2812
3. ip.proto: 0.1951
4. tcp.srcport: 0.0648
5. tcp.dstport: 0.0592
6. icmp.type: 0.0080
7. tcp.flags.syn: 0.0021
8. frame.time_delta: 0.0008
9. icmp.code: 0.0000

Model saved successfully as 'multiclass_ids_model.joblib'
```

Evaluation Metrics and Comparative Results

This section presents the performance evaluation of the Random Forest model using standard classification metrics, verifying its effectiveness in identifying the three distinct traffic classes (Normal, ICMP, and SYN Flood).

Evaluation Metrics Explained

The following metrics were used to thoroughly assess the model's performance, particularly in the context of network security:

- **Accuracy:** The overall percentage of all correctly classified packets (both attack and normal) out of the total test set.
- **Precision:** Measures the model's reliability when predicting an attack. It answers: Out of all packets the model predicted as an attack, how many were actually attacks? (Minimizing False Positives).
- **Recall (Sensitivity):** Measures the model's ability to detect all attacks. It answers: Out of all the actual attack packets, how many did the model correctly find? (Minimizing False Negatives).
- **F1-Score:** The harmonic mean of Precision and Recall, providing a balanced measure that is crucial when dealing with class imbalance.

Classification Results and Comparison

The model was tested on 108,143 unseen records (30% of the total dataset). The results confirmed a near-perfect ability to distinguish between the three highly separable classes:

- **Overall Accuracy:** The model achieved a displayed accuracy of 100.00% (due to rounding 0.9999537 to two decimal places). The true underlying accuracy was approximately 99.995%.

```

Classification Report (0=Normal, 1=ICMP, 2=SYN):

```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	651
1	1.00	1.00	1.00	42002
2	1.00	1.00	1.00	65490
accuracy			1.00	108143
macro avg	1.00	1.00	1.00	108143
weighted avg	1.00	1.00	1.00	108143

- **Comparison Analysis:** The perfect score of 1.00 across all metrics (Precision, Recall, F1-Score) for all three classes demonstrates that the model successfully created distinct boundaries between Normal traffic, ICMP Flood, and SYN Flood, validating the efficacy of the chosen features (ip.len, icmp.seq, ip.proto).

Analysis of Misclassifications (The Confusion Matrix)

While the overall accuracy was near-perfect, the Confusion Matrix revealed the exact 5 errors made by the model, providing crucial context:

```
Confusion Matrix:
[[ 648    1    2]
 [  0 4200    1]
 [  1    0 65489]]
```

- False Positives (FP): There were 3 FPs (1 Normal ->ICMP; 2 Normal ->SYN). These are benign packets flagged incorrectly as attacks.
- False Negatives (FN): There was 1 FN (1 SYN -> Normal). This represents a failed detection—an attack packet was missed and classified as safe traffic.
- Attack Misclassification: 1 packet was incorrectly identified as the wrong type of attack (ICMP ->SYN).

The existence of these 5 errors confirms that the model is not suffering from data leakage and that the 99.995% accuracy is the genuine, high-performing result achieved in the controlled lab environment.

Successful Deployment

The model was successfully deployed in a real-time monitor (live_test.py), demonstrating its practical capability to alert operators instantly upon detecting a change from safe status to a specific attack category

- **Normal live test**

```
[root@kali]~# ping 192.168.241.131
PING 192.168.241.131 (192.168.241.131) 56(84) bytes of data:
64 bytes from 192.168.241.131: icmp_seq=1 ttl=64 time=0.941 ms
64 bytes from 192.168.241.131: icmp_seq=2 ttl=64 time=0.805 ms
64 bytes from 192.168.241.131: icmp_seq=3 ttl=64 time=109 ms
64 bytes from 192.168.241.131: icmp_seq=4 ttl=64 time=342 ms
64 bytes from 192.168.241.131: icmp_seq=5 ttl=64 time=126 ms
64 bytes from 192.168.241.131: icmp_seq=6 ttl=64 time=3.32 ms
64 bytes from 192.168.241.131: icmp_seq=7 ttl=64 time=0.976 ms
64 bytes from 192.168.241.131: icmp_seq=8 ttl=64 time=1.00 ms
^C
— 192.168.241.131 ping statistics —
 8 packets transmitted, 8 received, 0% packet loss, time 7132ms
rtt min/avg/max/mdev = 0.805/72.940/341.501/112.812 ms
```

```
[root@kali]~# ftp 192.168.241.131
Connected to 192.168.241.131.
220 (vsFTPD 2.3.4)
Name (192.168.241.131:root): msfadmin
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
ftp>
ftp>
ftp>
ftp> exit
221 Goodbye.
```

[illegible]

- ICMP live test

```
(root@kali)-[~]
# sudo hping3 --icmp --flood -d 120 192.168.241.131
HPING 192.168.241.131 (eth0 192.168.241.131): icmp mode set, 28 headers + 120 data bytes
hping in flood mode, no replies will be shown
^C
— 192.168.241.131 hping statistic —
7155 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

```
[192.168.241.131] → [SAFE]: Normal Traffic
[192.168.241.129] → [SAFE]: Normal Traffic
[192.168.241.129] → [SAFE]: Normal Traffic
[192.168.241.131] → [SAFE]: Normal Traffic
[192.168.241.131] → [SAFE]: Normal Traffic
[192.168.241.129] → [SAFE]: Normal Traffic
[192.168.241.129] → [SAFE]: Normal Traffic
[192.168.241.131] → [SAFE]: Normal Traffic
[192.168.241.131] → [SAFE]: Normal Traffic
[192.168.241.129] → [SAFE]: Normal Traffic
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
```

- SYN live test

```
(root@kali)-[~]
# sudo hping3 -S --flood 192.168.241.131 -p 80
HPING 192.168.241.131 (eth0 192.168.241.131): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
— 192.168.241.131 hping statistic —
10944 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

```
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.131] → [*** DETECTED 1 ***]: ICMP Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.131] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.131] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.131] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.131] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.131] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.131] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.131] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
[192.168.241.129] → [*** DETECTED 2 ***]: SYN Flood Attack
```


Conclusion and Future Work

Conclusion

The project successfully achieved its objective of building and deploying a highly effective **Multi-Class Intrusion Detection System (IDS)** capable of distinguishing between Normal traffic, ICMP Flood, and SYN Flood attacks. The Random Forest Classifier achieved an exceptional effective accuracy of **99.995%** on the test set.

Key Success Points:

- **Effective Feature Engineering:** The chosen set of 9 features, particularly **ip.len** (packet size) and **icmp.seq** (sequencing), proved to be near-perfect discriminators, providing a clear separation between the three traffic patterns.
- **Model Performance:** The perfect metrics (Precision, Recall, F1-Score of **1.00** across all classes) validate that the model is robust and reliable at identifying the specific attack signatures learned in the lab environment.
- **Successful Deployment:** The model was successfully deployed in a real-time monitor (`live_test.py`), demonstrating its practical capability to alert operators instantly upon detecting a change from safe status to a specific attack category (Label 1 or Label 2).

Future Work and Project Expansion

To advance this research and prepare the IDS for a more realistic deployment, the following avenues are proposed for future work:

- **Expanded Protocol and Attack Scope:**

The model should be extended to include detection for **UDP Flood** attacks and common application-layer attacks (e.g., HTTP GET Floods). This would expand the classification task to a larger number of classes.

- **Model Optimization and Comparison:**

Compare the performance of the Random Forest Classifier against **Deep Learning** architectures, such as **Convolutional Neural Networks (CNNs)** or **Recurrent Neural Networks (RNNs/LSTMs)**, to determine if they can achieve better performance or generalization, particularly in handling the time-series nature of network traffic.